

Omar Alkhadra

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EDUCATION

University of Southern California <i>Master of Science in Applied Data Science (4.0/4.0)</i>	Los Angeles, CA Aug 2023 – May 2025
George Washington University <i>Bachelor of Science in Systems Engineering, Minor in Statistics (3.64/4.0)</i>	Washington, D.C. Aug 2019 – May 2023

EXPERIENCE

GEICO - Pricing <i>Data Scientist</i>	Chicago, IL Jun 2025 – Present
- Developed XGBoost and logistic regression models predicting attrition probabilities under rate changes and policy characteristics; 3× reduction in combined calibration error (Brier, ECE, log loss) against the incumbent baseline - Built stochastic simulation tool with interactive dashboard projecting policyholder attrition under rate-change scenarios, now used to inform pricing decisions across most state markets in production	
University of Southern California <i>Machine Learning Researcher</i>	Los Angeles, CA Dec 2024 – May 2025
- Co-authored paper published at ACM SIGSPATIAL '25 on probabilistic assignment of noisy GPS trajectories to candidate locations under spatial uncertainty - Reconstructed proprietary probabilistic baselines (SafeGraph's multi-stage KDE pipeline, Nishida's ILP formulation with spatiotemporal priors) as reproducible benchmarks	
AXA XL <i>Cyber Intern – Data Science</i>	Los Angeles, CA Jun 2024 – Aug 2024
- Initiated a claim-prediction project with the actuarial team, building multiple ML classifiers on 10,000+ policies to model filing risk from underwriter-assigned categoricals and historical claims data - Ran multivariate analysis and hypothesis testing to surface the features most associated with claim filing; identified key risk indicators and translated them into visualizations for underwriting and product teams	
George Washington University <i>Research Assistant</i>	Washington, D.C. Aug 2022 – May 2023
- Co-authored peer-reviewed paper in <i>Environment Systems and Decisions</i>; built dynamic inoperability input-output model (Leontief) to quantify economic losses across interdependent sectors facing drought - Conducted sensitivity analysis of policy intervention scenarios with expert-elicited parameters; validated model outputs against published economic loss estimates	

PROJECTS

Systematic Exploitation of Inefficient Prediction Markets
- Developed systematic trading strategy of temperature contracts across 20 cities with robust Kelly sizing, and autonomous daily retraining and execution; lifetime Sharpe of 2.0, with latest model seeing 37.4% RoR net of fees - Engineered XGBoost density model to price binary contracts; live forecasts beat market-implied prices with a 10.3% Brier skill score and 12× lower expected calibration error (0.006 vs. 0.072) across 10,757 market observations

SKILLS

- **Programming & ML:** Python (pandas, NumPy, SciPy, XGBoost, scikit-learn, PyTorch), SQL, R
- **Tools & Infrastructure:** Git, Linux, Jupyter, Snowflake, Apache Spark, REST APIs, AWS, Docker
- **Quantitative:** Probabilistic Calibration, Monte Carlo Methods, Stochastic Modeling, Time Series Analysis

PUBLICATIONS

- Saxena, N., Hsu, S., **Alkhadra, O.**, Shetty, M., Shahabi, C., & Horn, A. L. (2025). POIFormer: A Transformer-Based Framework for Accurate and Scalable Point-of-Interest Attribution. *SIGSPATIAL '25*, 607–619. ACM.
- Santos, J. R., **Alkhadra, O.**, Makrides, S., Huang, Y., & Menta, A. (2025). Drought impacts across economic sectors with policy analysis of mitigations. *Environment Systems and Decisions*.